



**PHILOMATH
NETWORKS PVT. LTD**



6-Days Hands-On Workshop on "Data Visualization Certification Training Course"

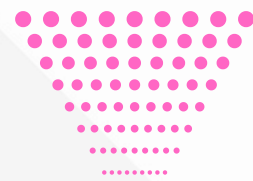
**Get ready to earn your Data
Visualization Certification with us!**

**Redefining Learning
Beyond Boundaries**



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About Philomath Networks



At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!



Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.

End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.

Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course Overview



▶ Course Objective

The Data Science & Analytics Certification Training Course is designed to equip participants with essential skills in data analysis, visualization, and machine learning. This course provides a hands-on learning experience using real-world datasets, covering key concepts such as data preprocessing, exploratory data analysis (EDA), and predictive modeling. Participants will gain proficiency in tools like Python, Power BI, and Tableau, enabling them to extract insights, build interactive dashboards, and make data-driven decisions. Through practical projects in business, healthcare, and finance, learners will develop industry-relevant expertise, preparing them for careers as data analysts, data scientists, or machine learning engineers.

▶ Course Outcome

✓ Understand Data Science

✓ Exploratory Data Analysis (EDA)

✓ Data Handling & Preprocessing

✓ Machine Learning Basics

✓ Data Visualization

✓ Hands-on Experience & Career Readiness

Training Modules

The Data Science & Analytics Certification Training is structured into six comprehensive modules, covering the complete data analysis workflow. The course begins with an Introduction to Data Science & Analytics, explaining key concepts, industry applications, and essential tools. Participants then explore Working with Data & Data Sources, where they learn about different data types, real-world datasets, and data collection methods from platforms like Kaggle and APIs. The training then moves into Data Preprocessing & Cleaning, focusing on handling missing values, data transformation, and feature engineering to ensure high-quality data for analysis.

Module 1: Introduction to Data Science & Analytics

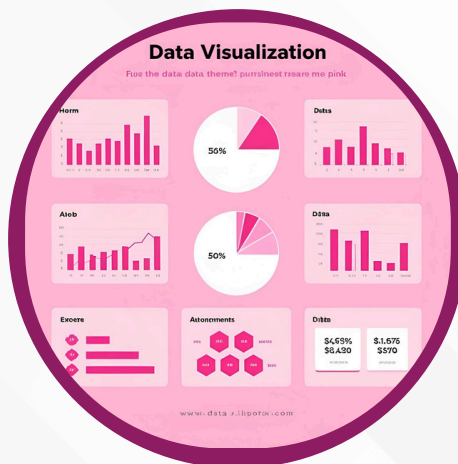
- Understanding Data Science and Analytics
- Importance of Data in Decision-Making Across Industries.
- Real-World Applications of Data Science.
- Essential Skills: Statistics, Programming, Data Visualization, and Machine Learning.
- Overview of Common Tools like Python, SQL, Power BI, Tableau, and Excel.
- Setting Up the Learning Environment

Module 2: Working with Data & Data Sources

- Understanding Different Types of Data: Structured, Unstructured, and Semi-Structured.
- Introduction to Public Data Repositories: Kaggle, Government Portals, APIs.
- Importing a Real-World Dataset into Python (Pandas) or Power BI.
- Exploring Key Columns: Job Titles, Salaries, Experience Levels, Industry, and Location.
- Understanding Data Quality & Identifying Potential Issues.
- Basic Data Cleaning Techniques: Handling Missing Values & Duplicates.

Module 3: Data Preprocessing & Cleaning

- Identifying and Handling Missing Data (Mean, Median, Mode Imputation).
- Removing Duplicates and Handling Data Inconsistencies.
- Data Transformation Techniques: Normalization, Standardization, and Encoding.
- Feature Engineering: Extracting Insights & Creating New Variables.
- Splitting Data for Training and Testing in Predictive Models.
- Hands-on Practice: Cleaning and Preparing the Job Dataset for Analysis.



Training Modules

The next phase covers Data Visualization & Exploratory Data Analysis (EDA), where learners build charts and interactive dashboards using Matplotlib, Seaborn, Power BI, or Tableau. This is followed by an Introduction to Machine Learning & Predictive Analytics, where participants learn about regression, decision trees, and clustering while building a salary prediction model. Finally, the Career Paths & Real-World Projects module helps learners apply their skills through industry-relevant projects, portfolio building, and career guidance to prepare for roles in data science and analytics.

Module 4: Data Visualization & Exploratory Data Analysis (EDA)

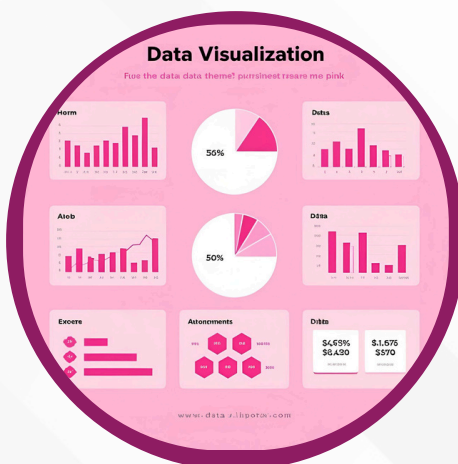
- Importance of Data Visualization in Decision-Making.
- Creating Basic Charts (Bar, Line, Pie) using Matplotlib & Seaborn.
- Building Interactive Dashboards with Power BI or Tableau.
- Identifying Patterns, Trends, and Outliers in Data.
- Detecting Correlations & Relationships Between Variables.
- Hands-on: Visualizing Job Market Trends and Salary Distribution

Module 5: Introduction to Machine Learning & Predictive Analytics

- Understanding Machine Learning: Supervised vs. Unsupervised Learning.
- Overview of Key Algorithms: Linear Regression, Decision Trees, and Clustering.
- Hands-on: Building a Salary Prediction Model Using Linear Regression.
- Evaluating Model Performance: Accuracy, Precision, and Recall.
- Feature Importance: Identifying Which Factors Affect Salaries the Most.
- Hands-on Practice: Experimenting with Different ML Models on Job Dataset.

Module 6: Real-World Projects

- Career Roles – Data Scientist, Data Analyst, ML Engineer.
- Tools & Certifications – Python, SQL, Power BI, Tableau.
- Industry Projects – Job market, e-commerce, sales, healthcare.
- Predictive Analytics – Forecast trends, data-driven decisions.
- Resume & Portfolio – Build a strong data science profile.
- Q&A & Certification – Career guidance, project showcase, certificate.



Training Schedule & Format



The training schedule is structured as a 6-day intensive workshop, providing a hands-on learning experience in Data Science and Analytics. Each day focuses on key concepts, starting with fundamental topics like data handling and progressing to advanced areas such as machine learning and real-world case studies. The workshop includes interactive sessions, data analysis exercises, and visualization techniques to reinforce learning. Participants will work on hands-on projects and practical assignments daily, ensuring real-world application of concepts. The final day is dedicated to career guidance, industry tools, and certification, equipping learners with job-ready data science skills.

Day	Module	Focus Area	Duration
Day 1	Introduction to Data Science & Analytics	Key Concepts, Applications, Tools	5hrs
Day 2	Working with Data & Data Sources	Data Collection	5hrs
Day 3	Data Preprocessing & Cleaning	Cleaning, Transformation	5hrs
Day 4	Data Visualization & Exploratory Data Analysis (EDA)	Charts, Dashboards, Trend Analysis	5hrs
Day 5	Introduction to Machine Learning & Predictive Analytics	Regression, , Model Building	5hrs
Day 6	Real-World Projects	Job Market, Sales, Healthcare Analysis	5hrs

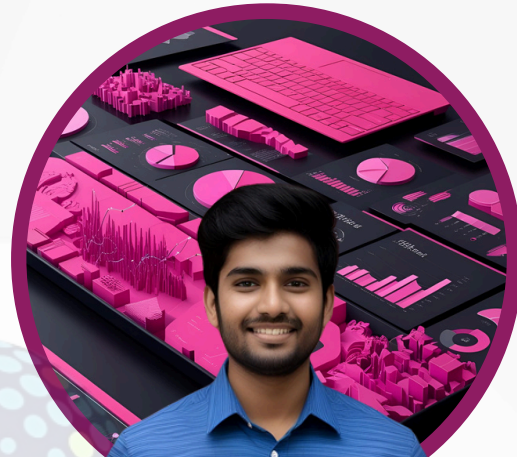
Through hands-on exercises and real-world case studies, participants will develop essential data science skills, explore best practices, and tackle analytical challenges. The workshop includes live demonstrations, guided projects, and interactive activities to enhance learning. A collaborative format encourages discussions, problem-solving, and peer learning. By the end, attendees will have a strong foundation in data analysis, visualization, and machine learning, enabling them to extract insights, make data-driven decisions, and build industry-relevant projects.

How to Get Involved



Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact Information

For any questions about the training program, please reach out to:

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